# Why Precision Livestock Farming will create a more sustainable livestock sector

## What we can learn from big data

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4th International Huvepharma Seminar
Today's challenges for optimal nutrition & health of high performing animals
6-8 December 2022, Penang, Malaysia

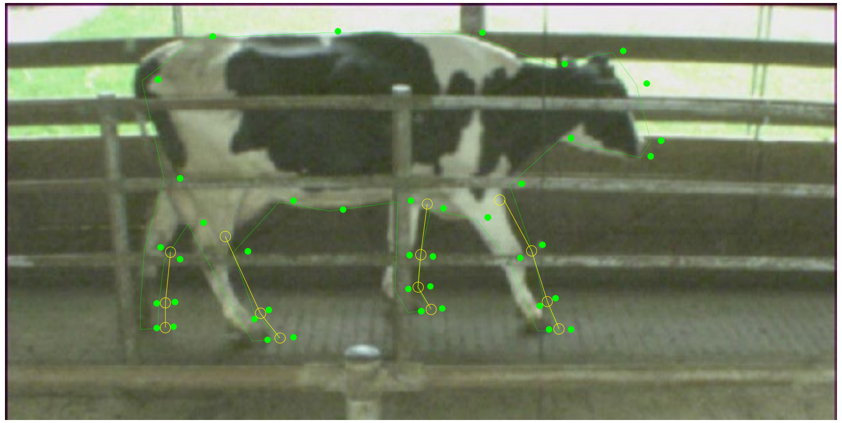
# Outline

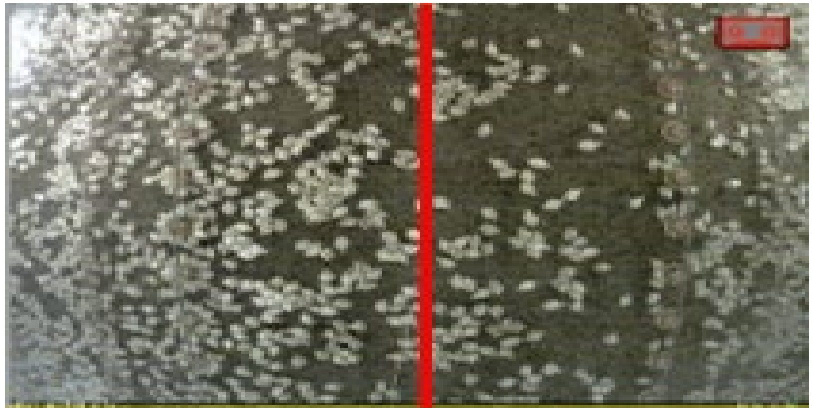
- What is Precision Livestock Farming?
- Real-time monitoring by Precision Livestock Farming (PLF)
- Predictive-based nutrition control by PLF technology
- Conclusions

-> (5 reasons why PLF will create a more sustainable livestock sector)

# Definition of Precision Livestock Farming (PLF)

# Precision Livestock Farming is:





A tool for management of livestock by continuous automated real-time monitoring of production/reproduction, health and welfare  and environmental impact.

*Reason 1: the worldwide livestock sector is facing huge challenges… time for changes*

# Challenges for the livestock production

- Over 70 billion animals slaughtered every year while increase of worldwide demand for animal products with up to 70% by 2050 (FAO)?
- **Health:** Relationship with human health (e.g., Covid-19, ASF)

⬆

- **Animal welfare** (e.g. EU)

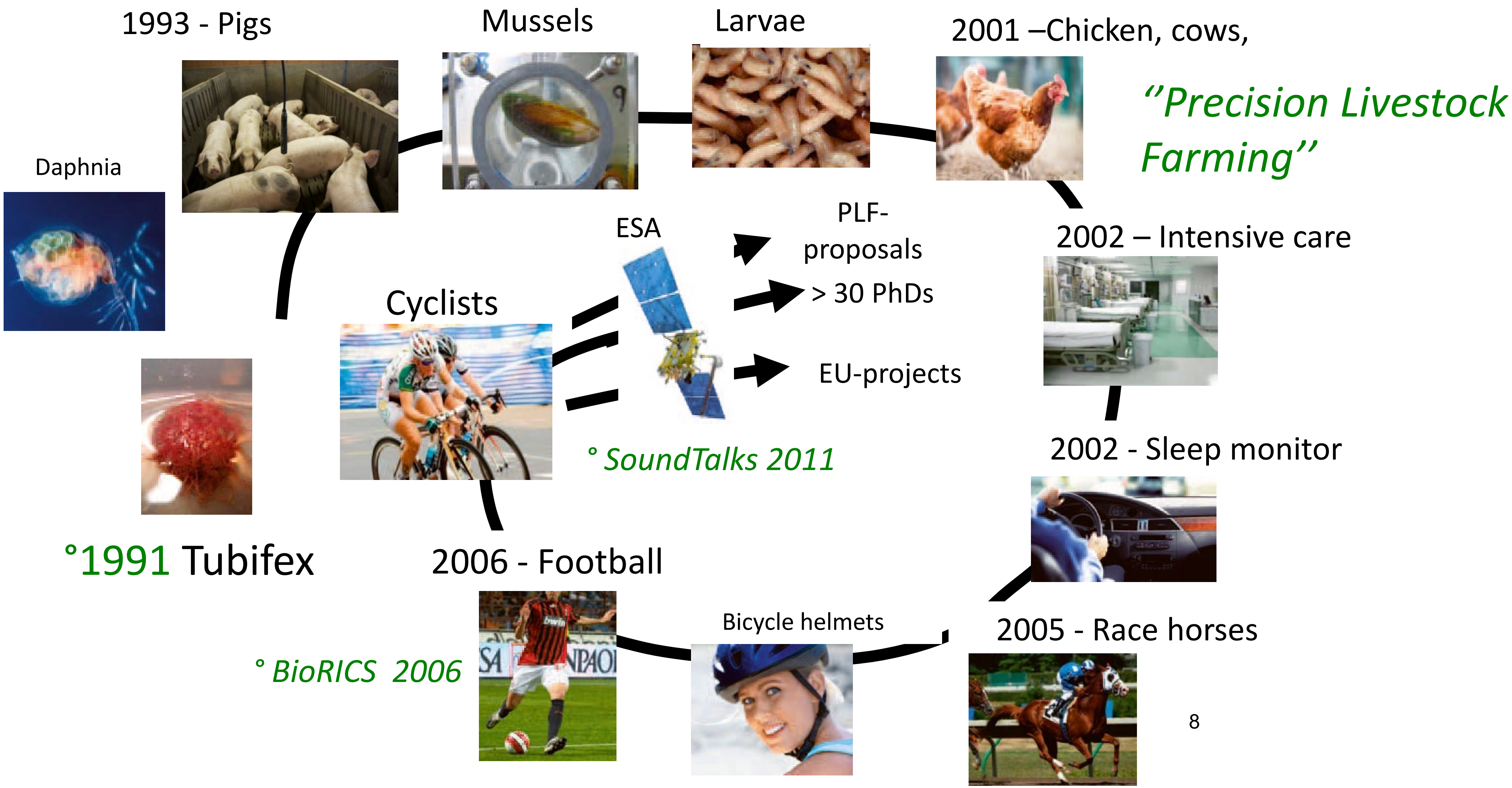
⬇

- **Environmental Issues**
- **Social importance**
- **Economic importance** including Valorisation of knowledge

**25–27 September 2015     UN HQ, New York City, United States of America**

193 Member States of the UN agree:
Transforming  Our World: 2030 Agenda for Sustainable Development
A Declaration with **17 Sustainable Development Goals    ->    9 Livestock connected SDG's**

# Reason 2: PLF has a scientific basis

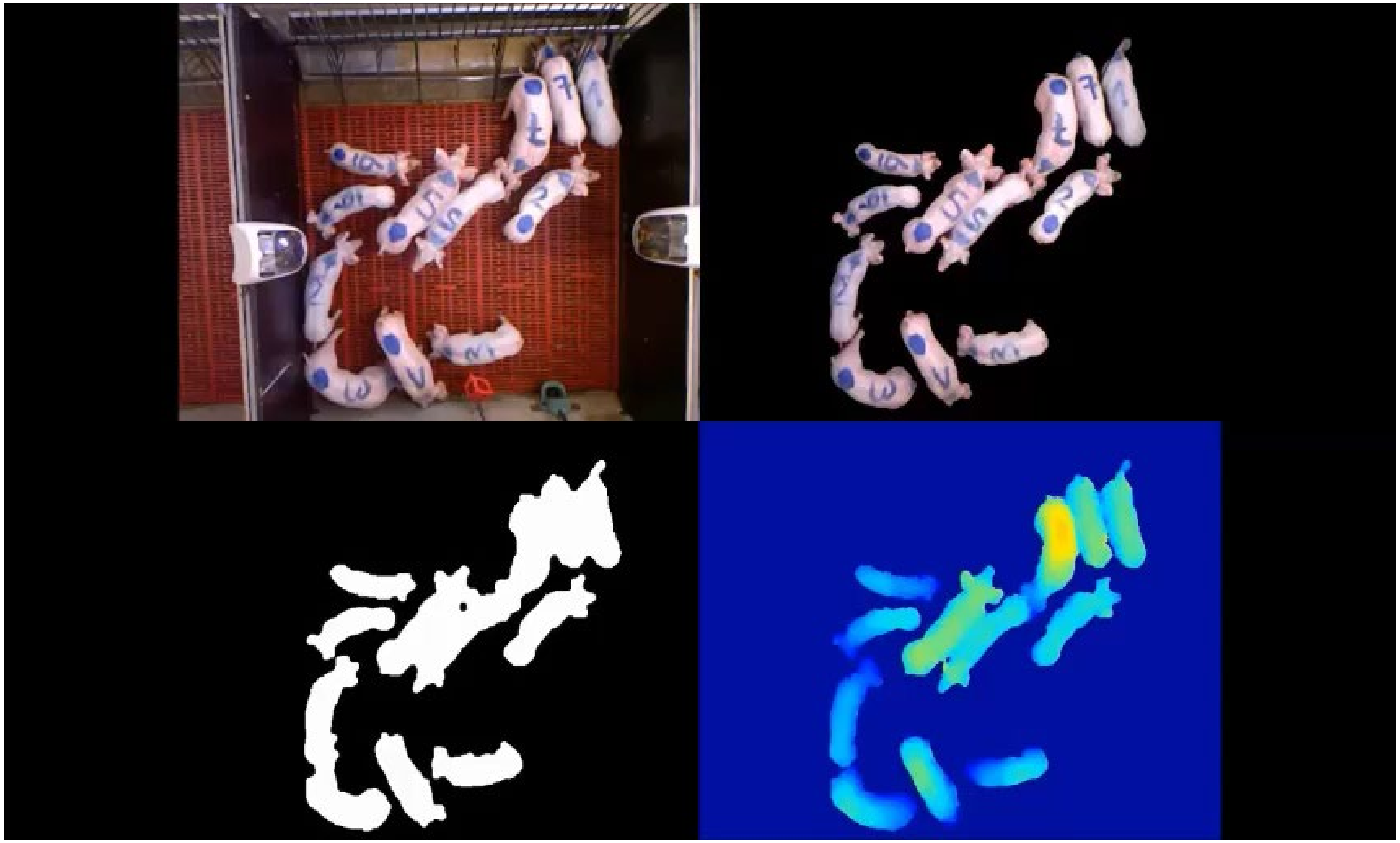



2022: 10th ECPLF Conference -> over 5000 pages of papers

## *Reason 3: PLF offers continuous objective monitoring of livestock in an automated way*

# Aggression monitor (2014)

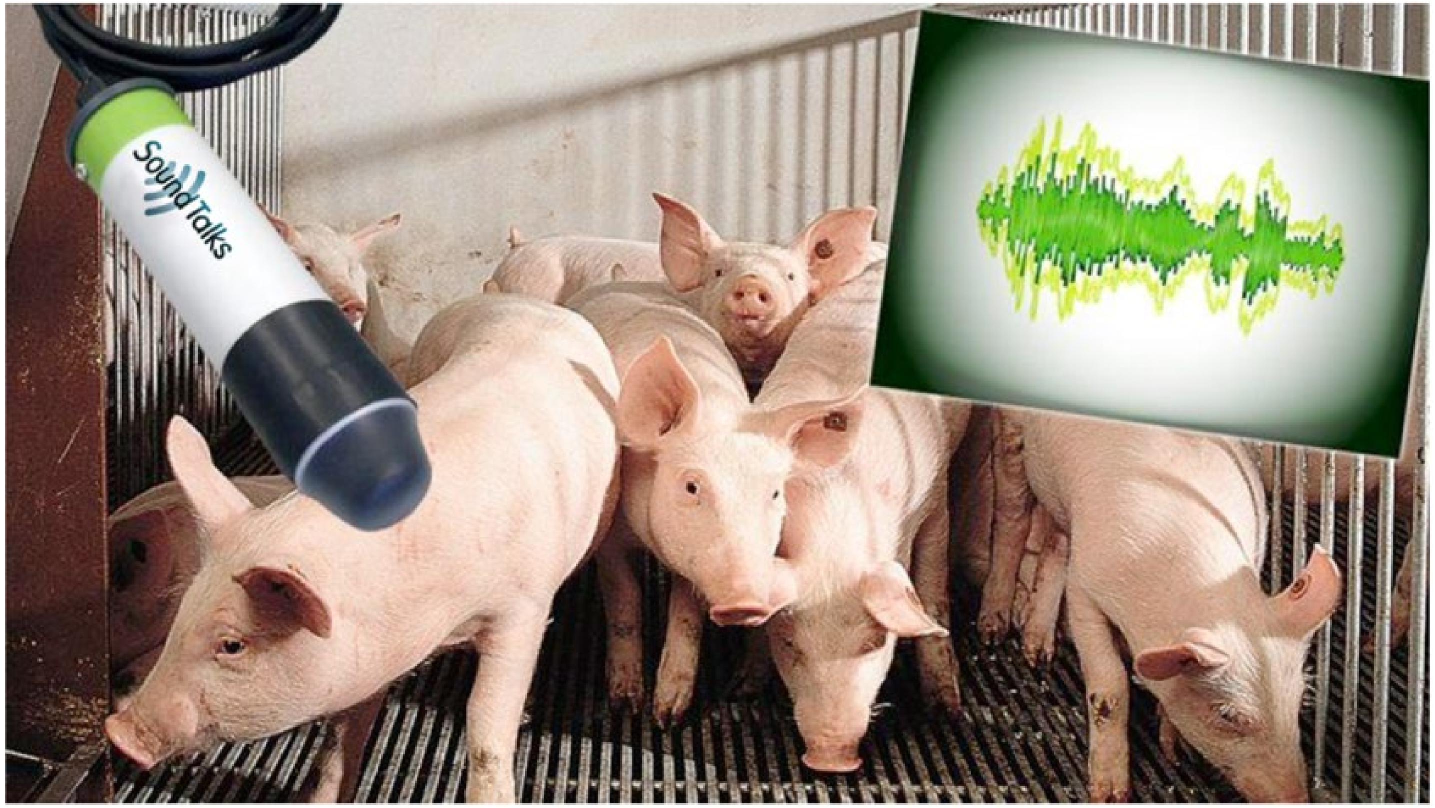
i.c.w. Umil (Italy), TIHO (Germany), Fancom (the Netherlands)



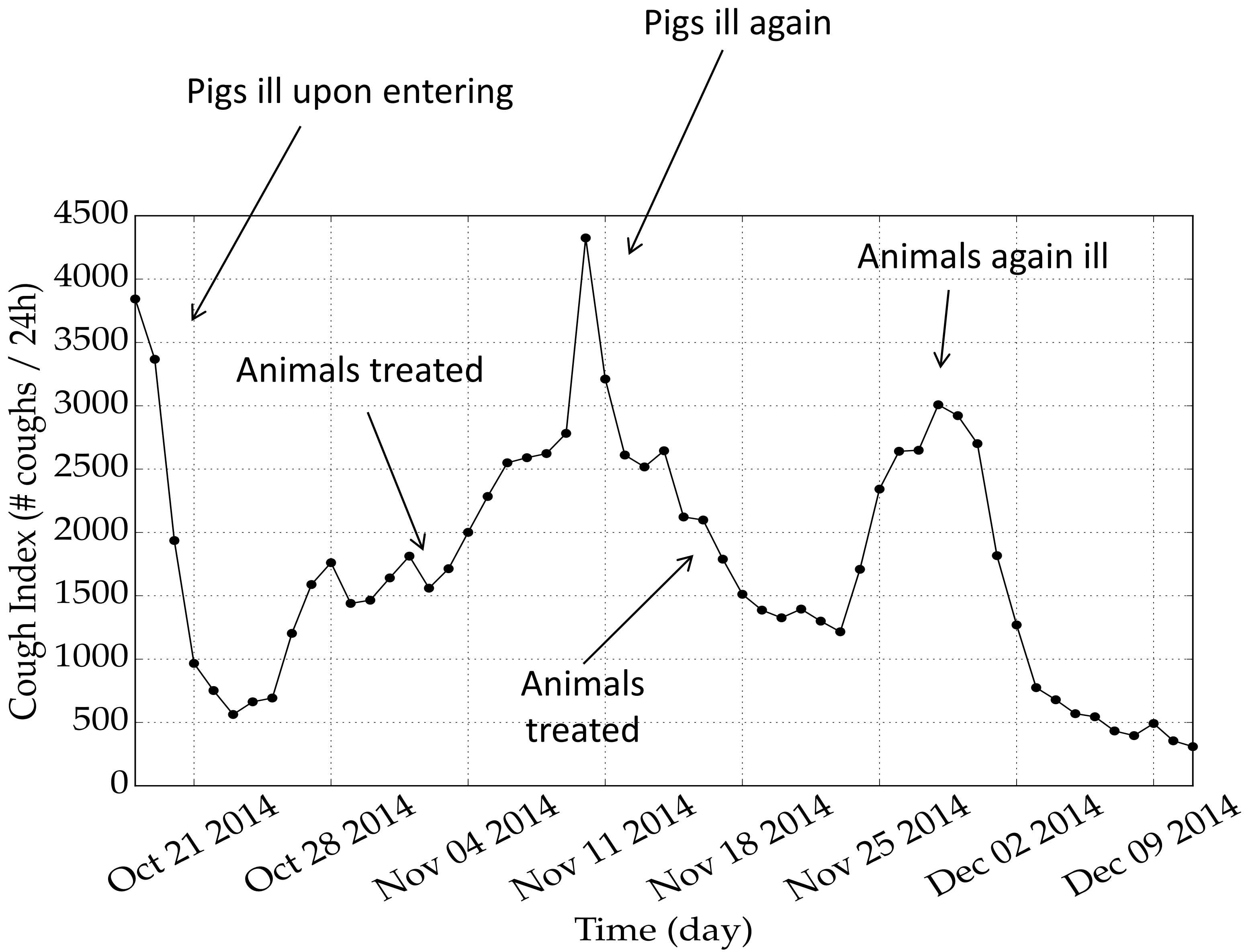
Accuracy of 89 %

# “Small” animals: Infection Monitoring by On-line Sound Analysis

i.c.w. University of Milan, SoundTalks NV, Boehringer-Ingelheim



# Results EU-PLF (2016)

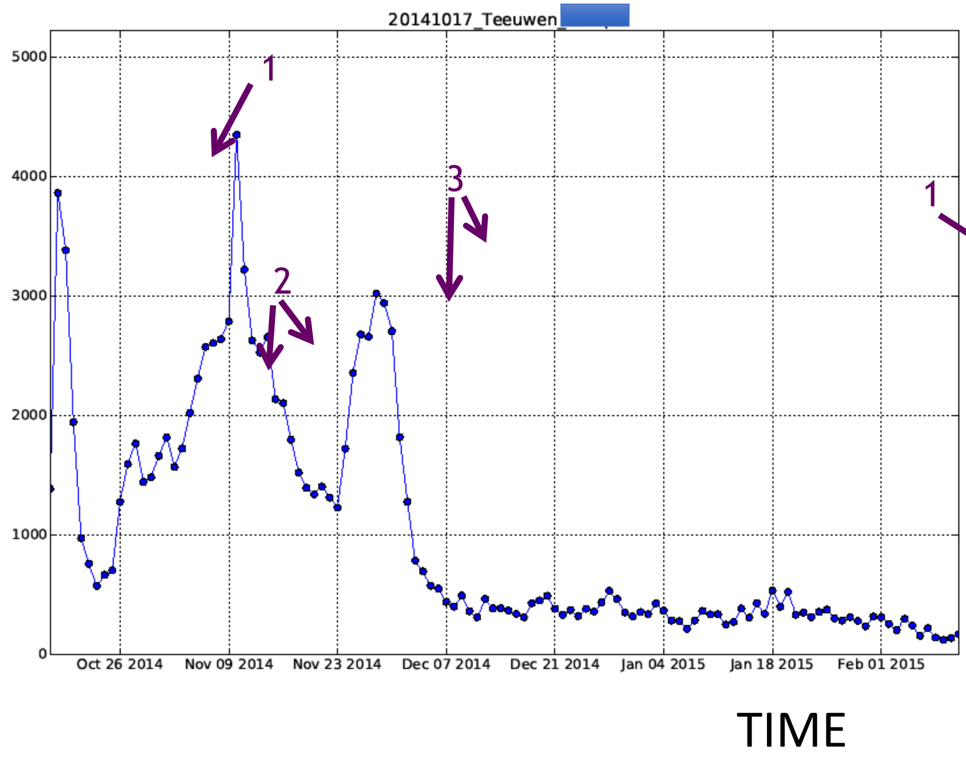




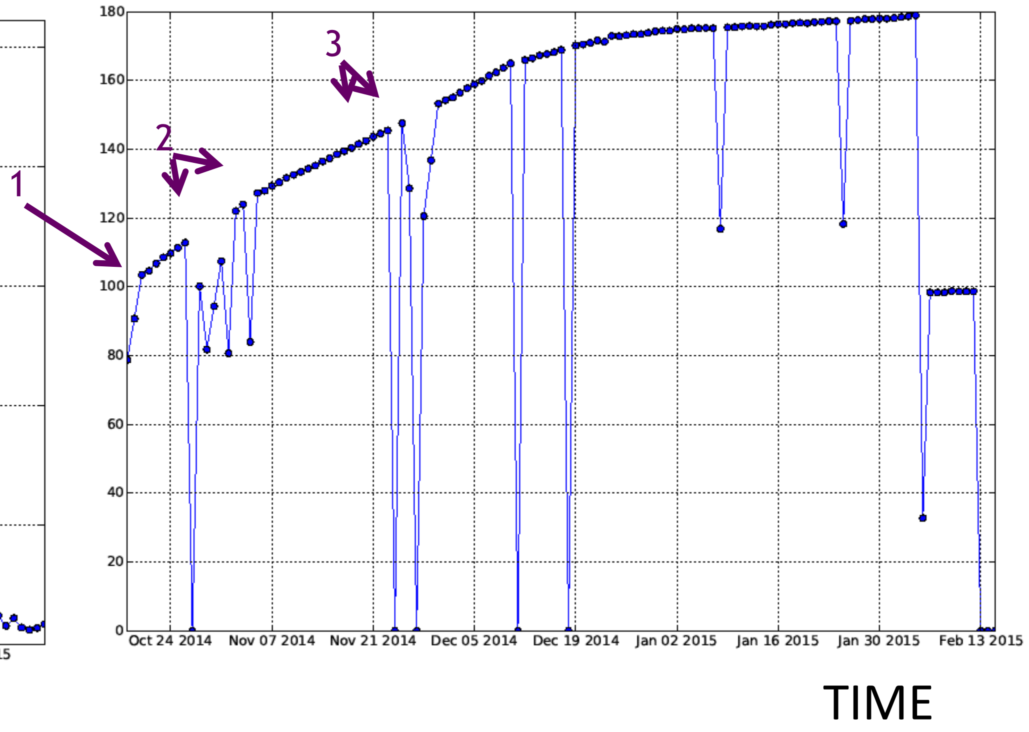

(SoundTalks & Boehringer Ingelheim)

# Feed delivery vs respiratory distress

Cough index




Feed delivered



*Reason 4: PLF will allow continuous monitoring of the core equation of the livestock sector*

# Fundamentals of the biological process in livestock: Transfer feed energy into animal product

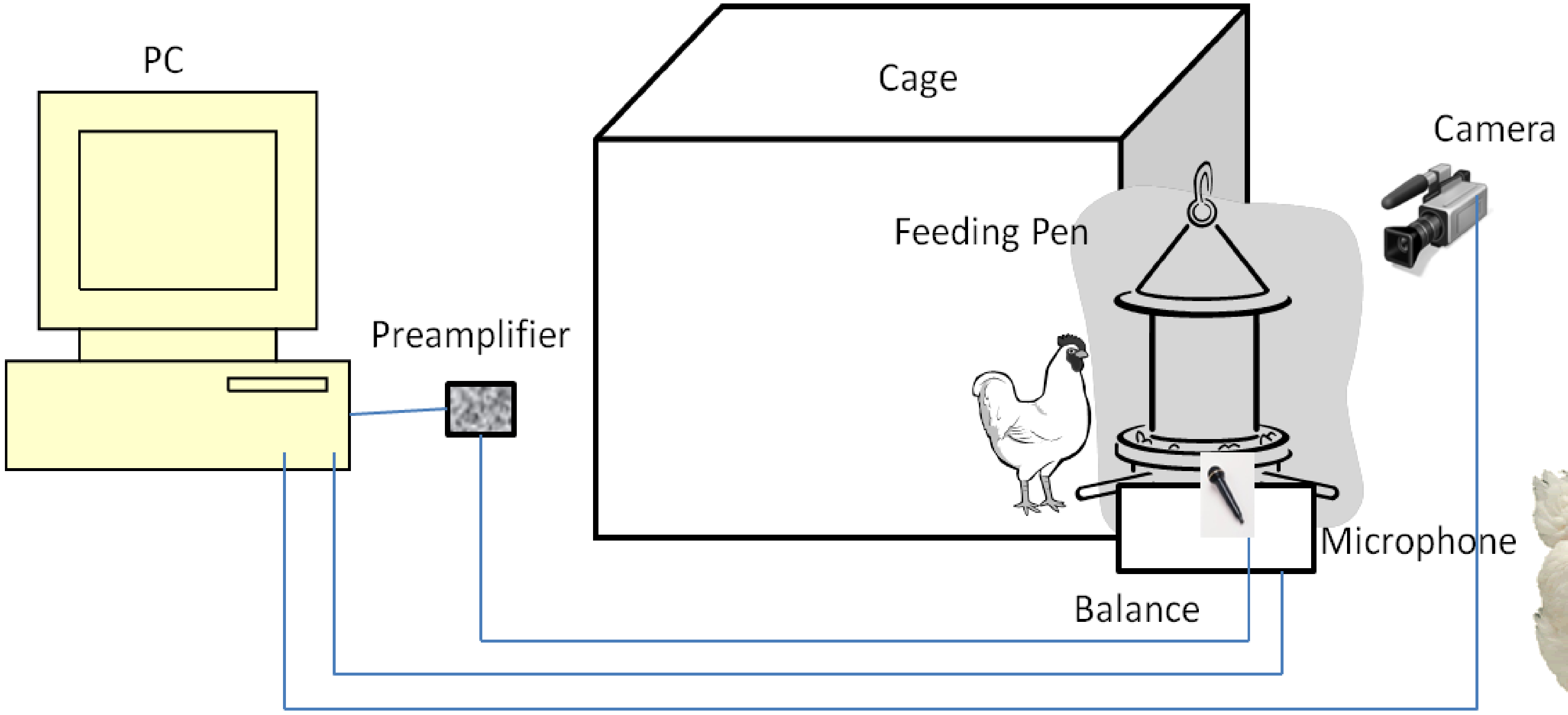
$$E_{tot} = E_{bas\ Met} + E_{activ} + E_{therm} + E_{mental} + \color{red}{E_{production}} + \xi$$

Feed intake   & Immune   Animal welfare   Meat, milk, eggs, animals

# Monitoring of feed intake of broilers by sound technology

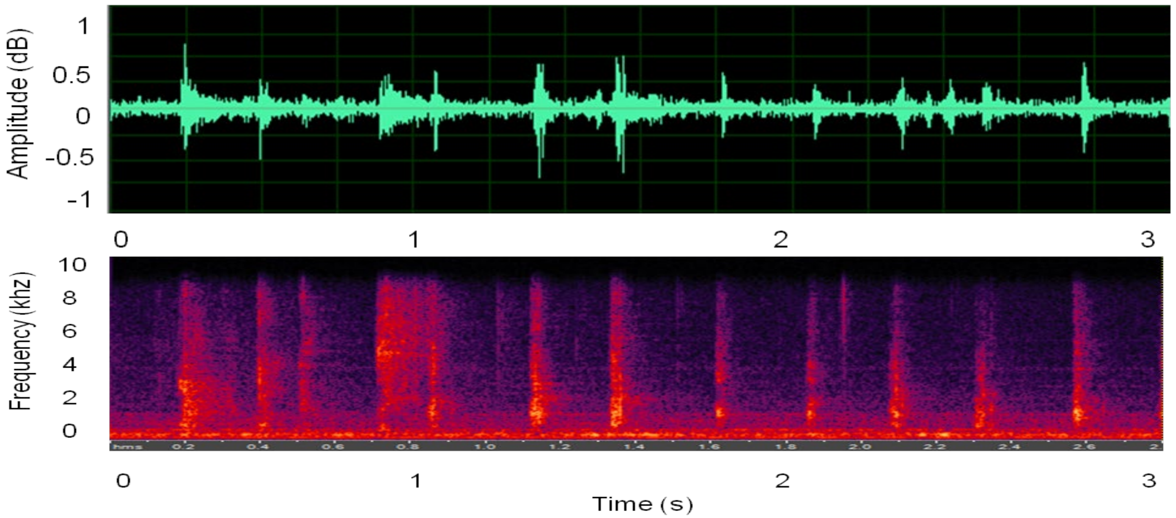
# MATERIALS and METHOD

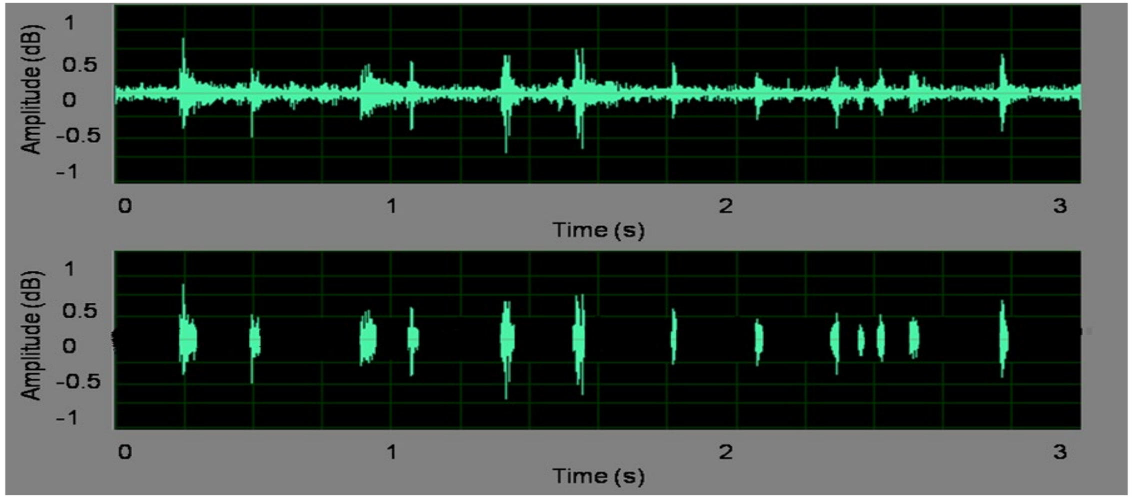
- 12 male, 28 days old, Ross 308 broilers
- 3 repetition for each chicken.
- 36 experiments in total.




# Real-time monitoring of feed intake

Play



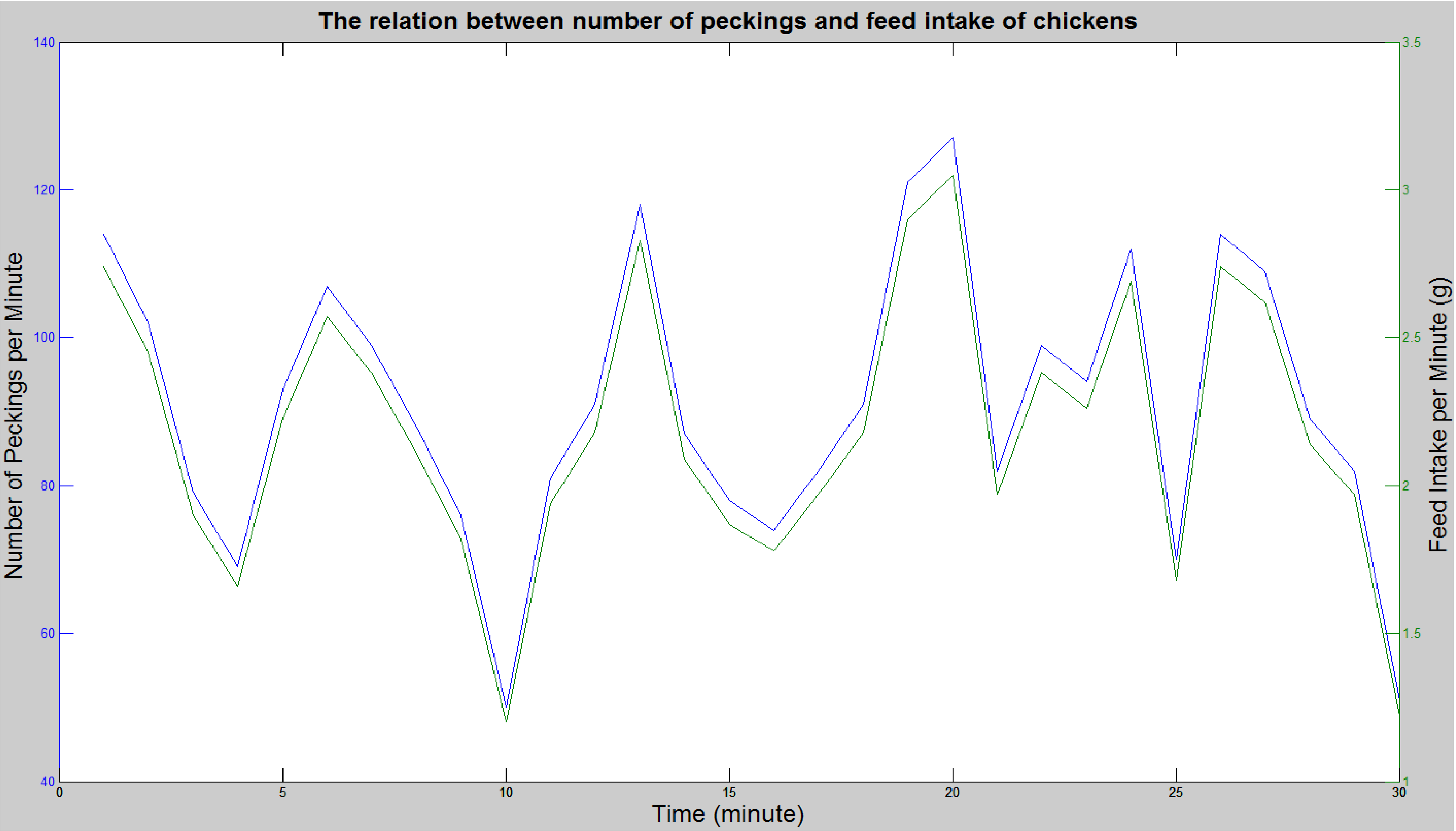




Play

Continuous recording of sounds (top) and individual pecking sounds (bottom) as extracted by the algorithm.

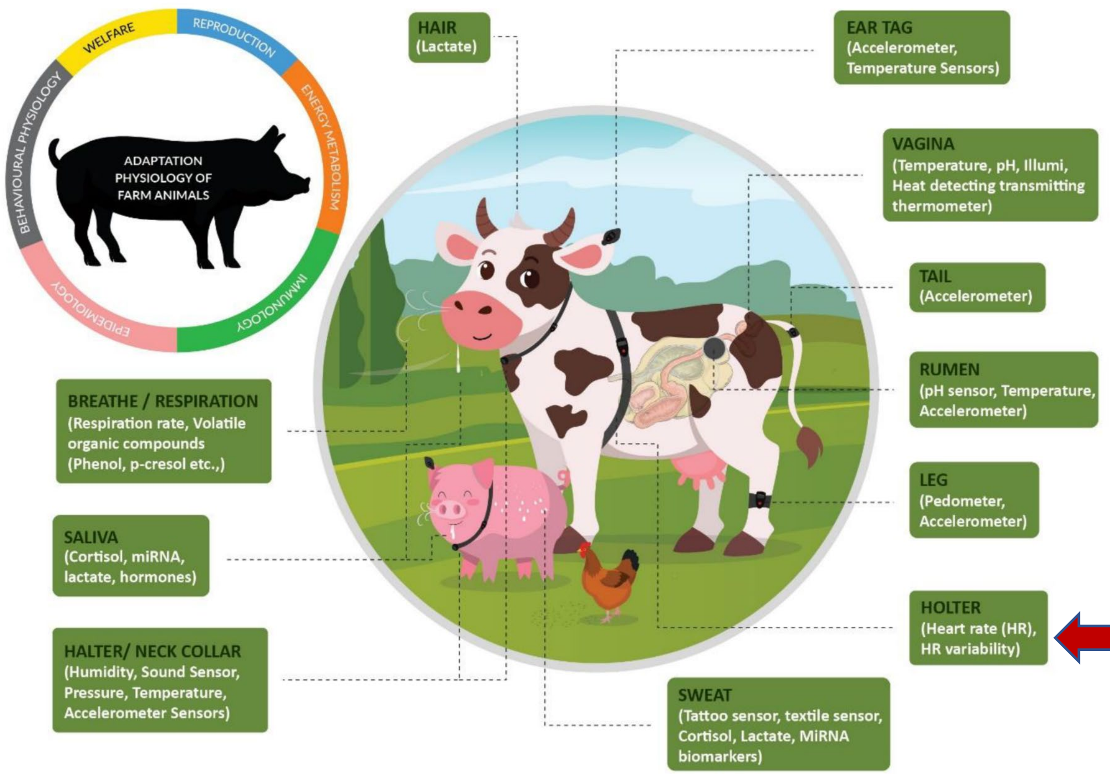
# Results of monitoring feed intake

| Data Set | Number of pecking (Algorithm) | Number of pecking ( Video Labelling) | Accuracy of Algorithm (%) | True Positive | False Positive |
|---|---|---|---|---|---|
| 1 | 113 | 105 | 93 | 105 | 8 |
| 2 | 99 | 95 | 96 | 95 | 4 |
| 3 | 109 | 106 | 98 | 106 | 3 |
| 34 | 107 | 101 | 94 | 101 | 6 |
| 35 | 98 | 91 | 93 | 91 | 7 |
| 36 | 95 | 88 | 92 | 88 | 7 |
| Total-Average | 3707 | 3447 | 93 | 3447 | 7 |


The relation between number of peckings and feed intake of chickens
Number of Peckings per Minute
Feed Intake per Minute (g)
Time (minute)
140
120
100
80
60
40
3.5
3
2.5
2
1.5
1
0
5
10
15
20
25
30

# Monitoring the **basal metabolism**

Transforming the Adaptation Physiology of Farm Animals through Sensors.
Suresh Neethirajan, 2020. Animals 2020, 10, 1512, 1-24.

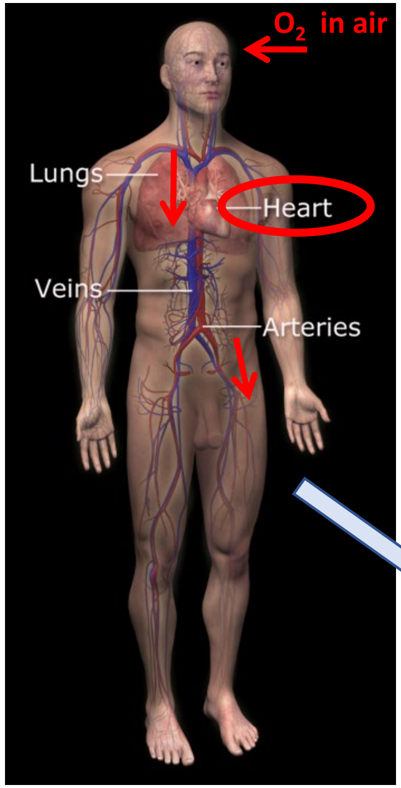



Heart Rate

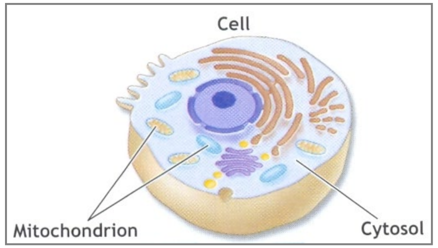
$$E_{tot} = E_{bas\ Met} + E_{activ} + E_{therm} + E_{mental} + E_{production} + \xi$$

# Real-time monitoring of the Aerobic metabolic energy production in the body

Heart rate is a measure for metabolic energy production




$O_2$




ATP

Aerobic energy

**Heart rate: & Activity**

**Algorithm**

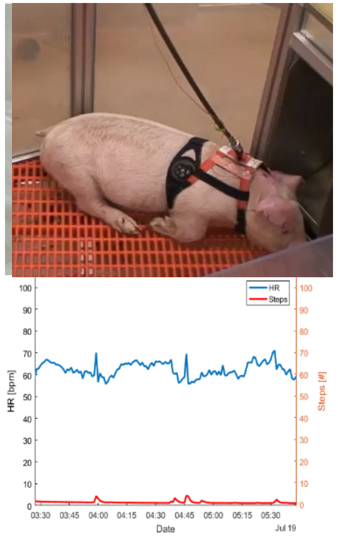
**Where is the energy used for:**

- **maintenance,**
- **activity,**
- **thermal,**
- **mental**
- **immune system**

# Monitoring metabolic energy in basal, movement and mental term

HR

Movement




Time

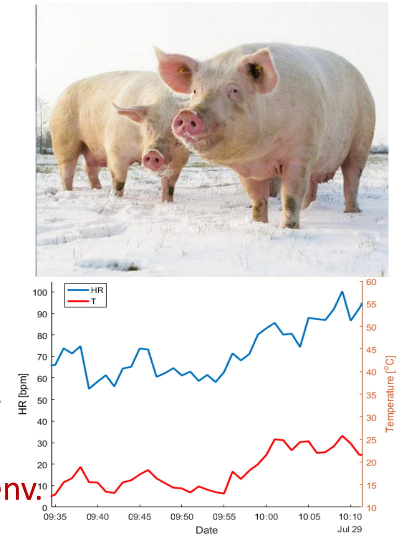
HR

Mov.


Time

HR

T env.



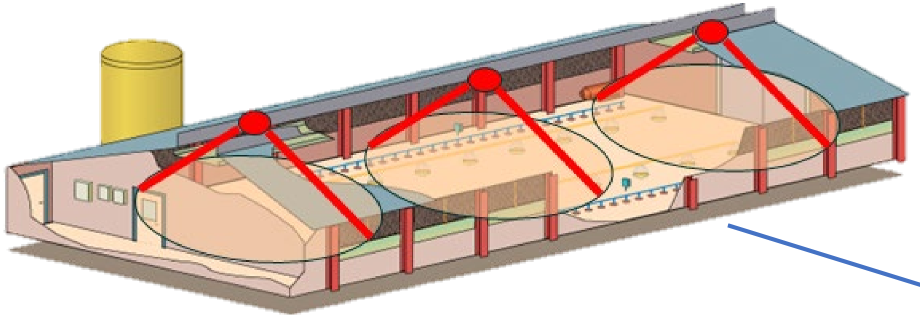

Time

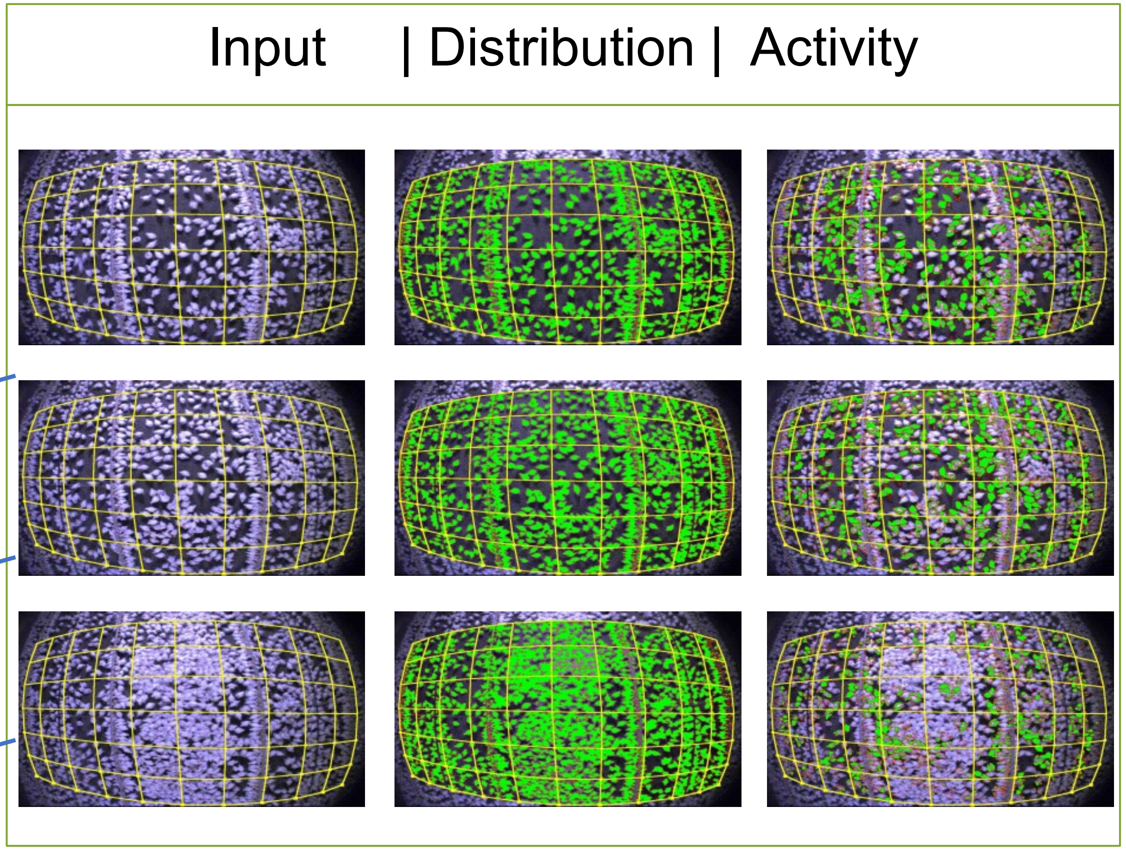
**HRtotal** = **HR basal** + **HR movement** + **HR thermal** + **HR mental**

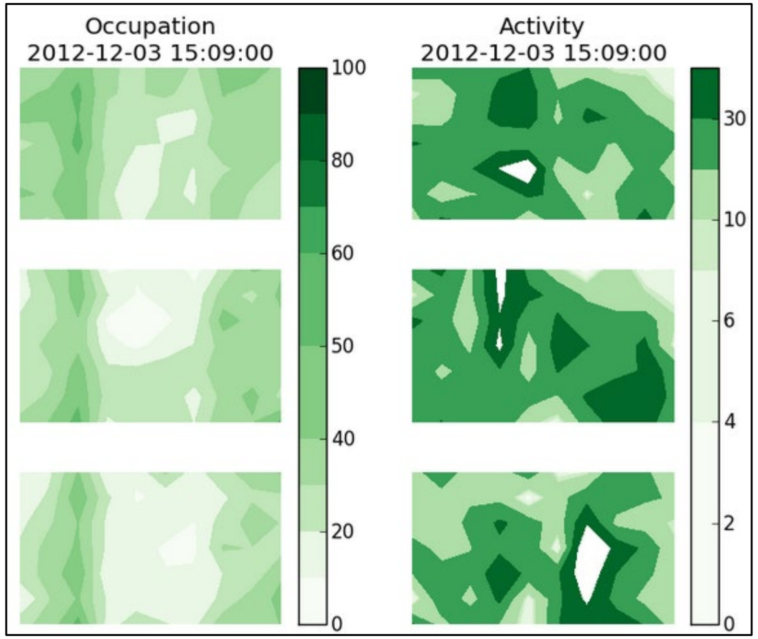
# Automated monitoring of the **Activity**

$$E_{tot} = E_{bas\ Met} + E_{activ} + E_{therm} + E_{mental} + E_{production} + \xi$$

# eYeNamic for broilers





# Monitoring the metabolic energy for **animal welfare**

-Is Continuous Heart Rate Monitoring of Livestock a Dream or is it Realistic? A Review.
Nie LW., Berckmans D., Wang CY., Li BM. 2020, SENSORS, 20, 8.

- An Approach towards Motion-Tolerant PPG-Based Algorithm for Real-Time Heart Rate Monitoring of Moving Pigs.
Youssef A.; Fernandez A.P.; Wassermann L., ; Biernot S.; Wittauer E.M. Bleich A., Hartung J. Berckmans D., Norton T. 2020. SENSORS, 20, 15.

- Non-Invasive PPG-Based System for Continuous Heart Rate Monitoring of Incubated Avian Embryo.
Youssef A., Berckmans D., Tomas N. 2020. Sensors 2020, 20, 4560, 1- 17.

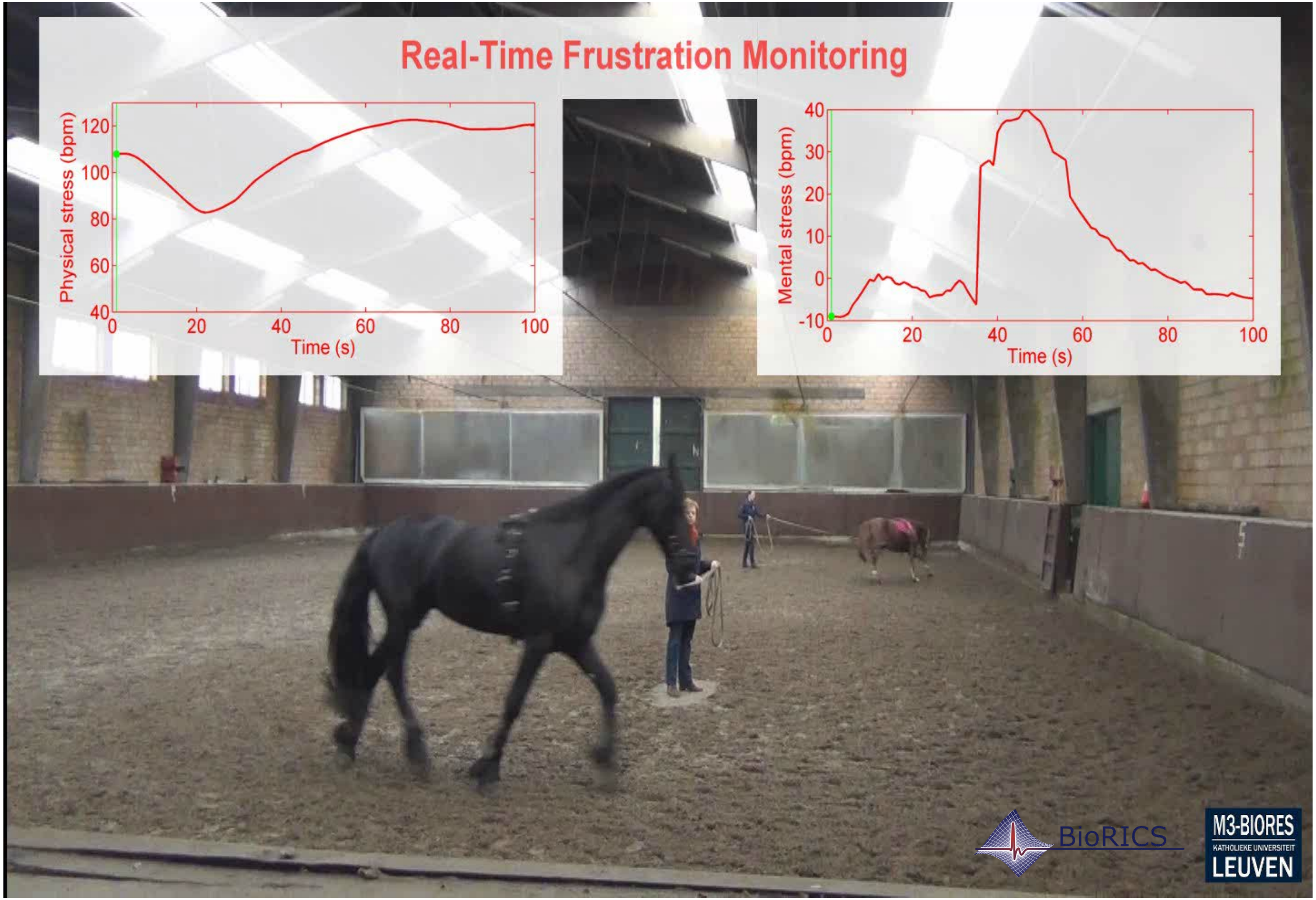
$$E_{tot} = E_{bas\ Met} + E_{activ} + E_{therm} + E_{mental} + E_{production} + \xi$$

Feed intake

Animal welfare

Meat, milk, eggs, animals

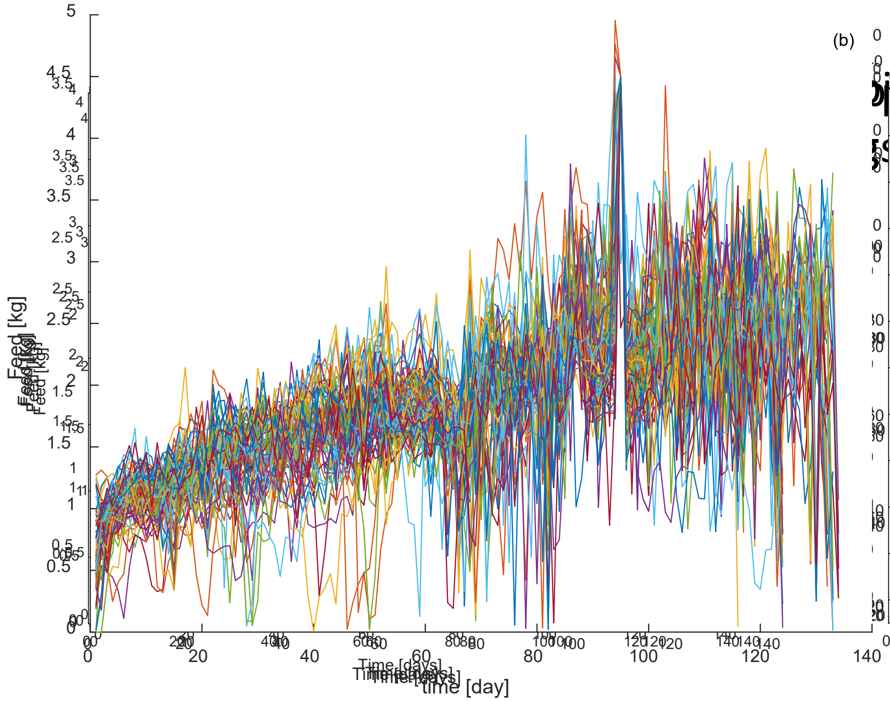
# Real-Time Animal welfare monitoring by physiological variables



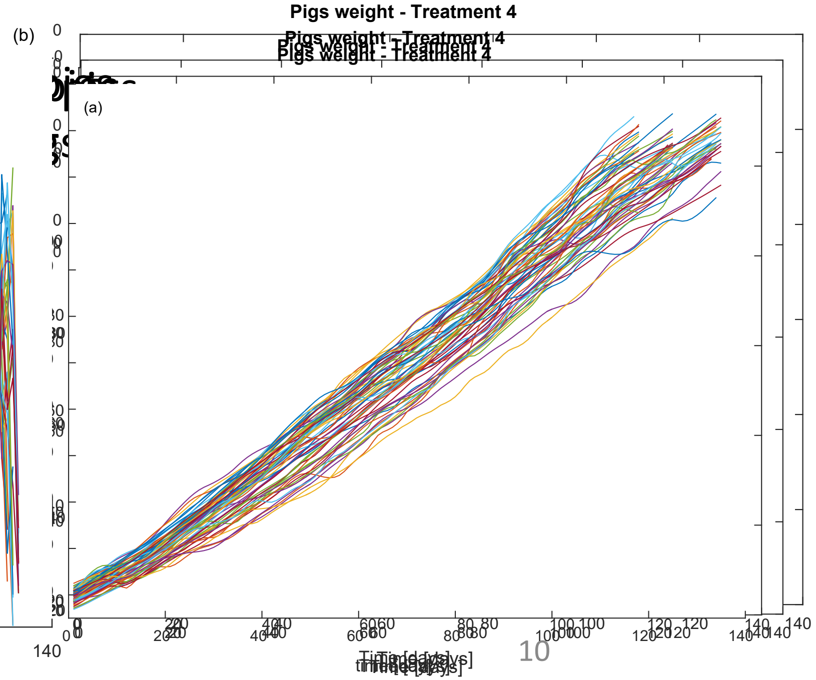
Real-Time Frustration Monitoring
Physical stress (bpm)
120
100
80
60
40
0
20
40
60
80
100
Time (s)
Mental stress (bpm)
40
30
20
10
0
-10
0
20
40
60
80
100
Time (s)
BioRICS
M3-BIORES
KATHOLIEKE UNIVERSITEIT
LEUVEN

# Continuous monitoring of **production** (Pig Weight) by camera

## Feed supply

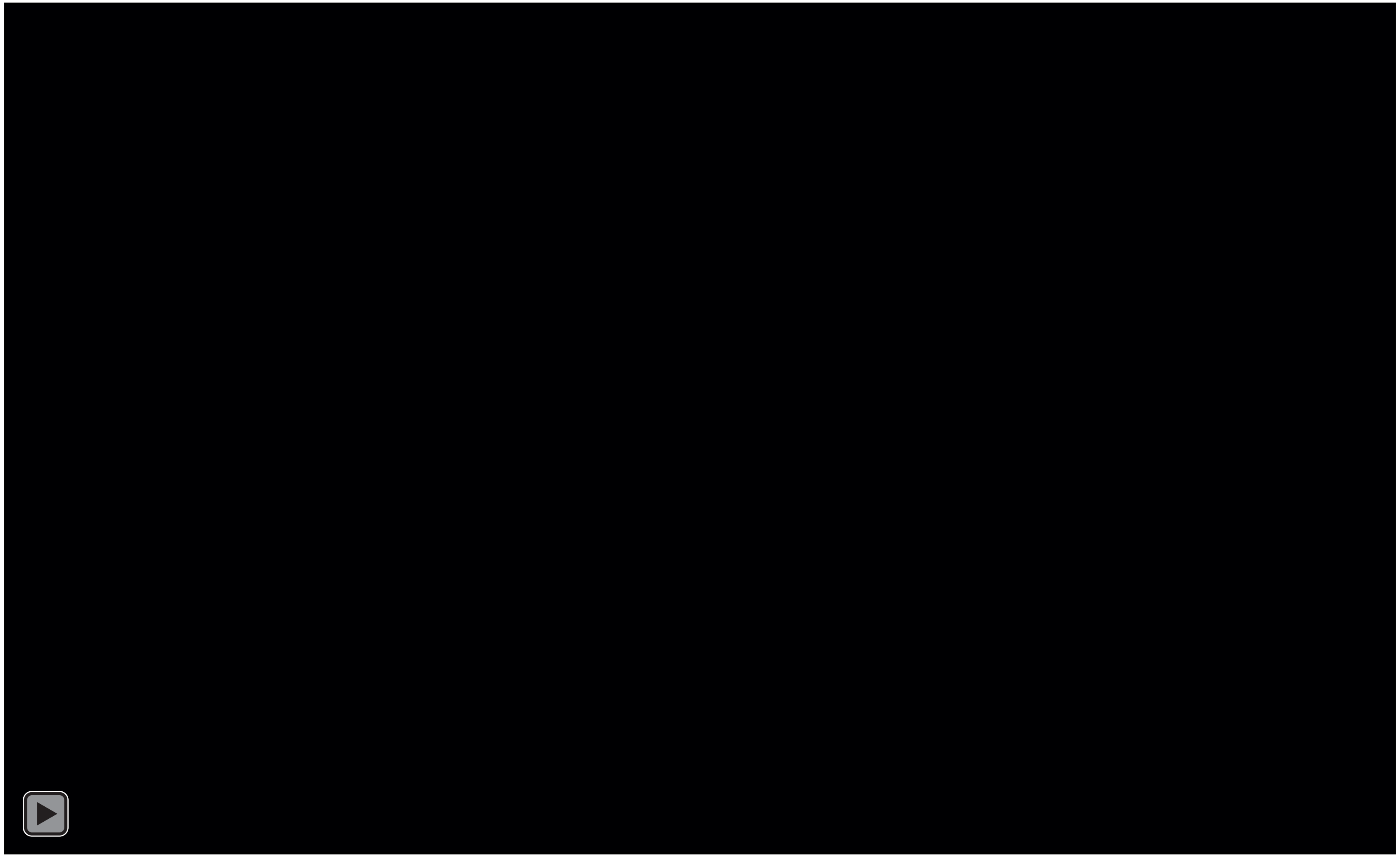



## Weight



# Continuous weight monitoring

i.c.w. Fancom (the Netherlands)



Accuracy 0.82 kg

## *Reason 5: New technology will support <u>management</u> of the core equation of the livestock sector*

# PLF supports monitoring & controlling of each term of the energy equation

$$E_{tot} = E_{bas\ Met} + E_{activ} + E_{therm} + E_{mental} + \color{red}{E_{production}} + \xi$$

Feed intake

Animal welfare

Meat, milk, eggs, fibers, reproduction

# Predictive based control

# Model-based Control in other processes


# Principle of Predictive-based control

Position and balance on the board

Direction

MEASURE

2

MEASURE

1

PREDICTION

CORRECT THE INPUT

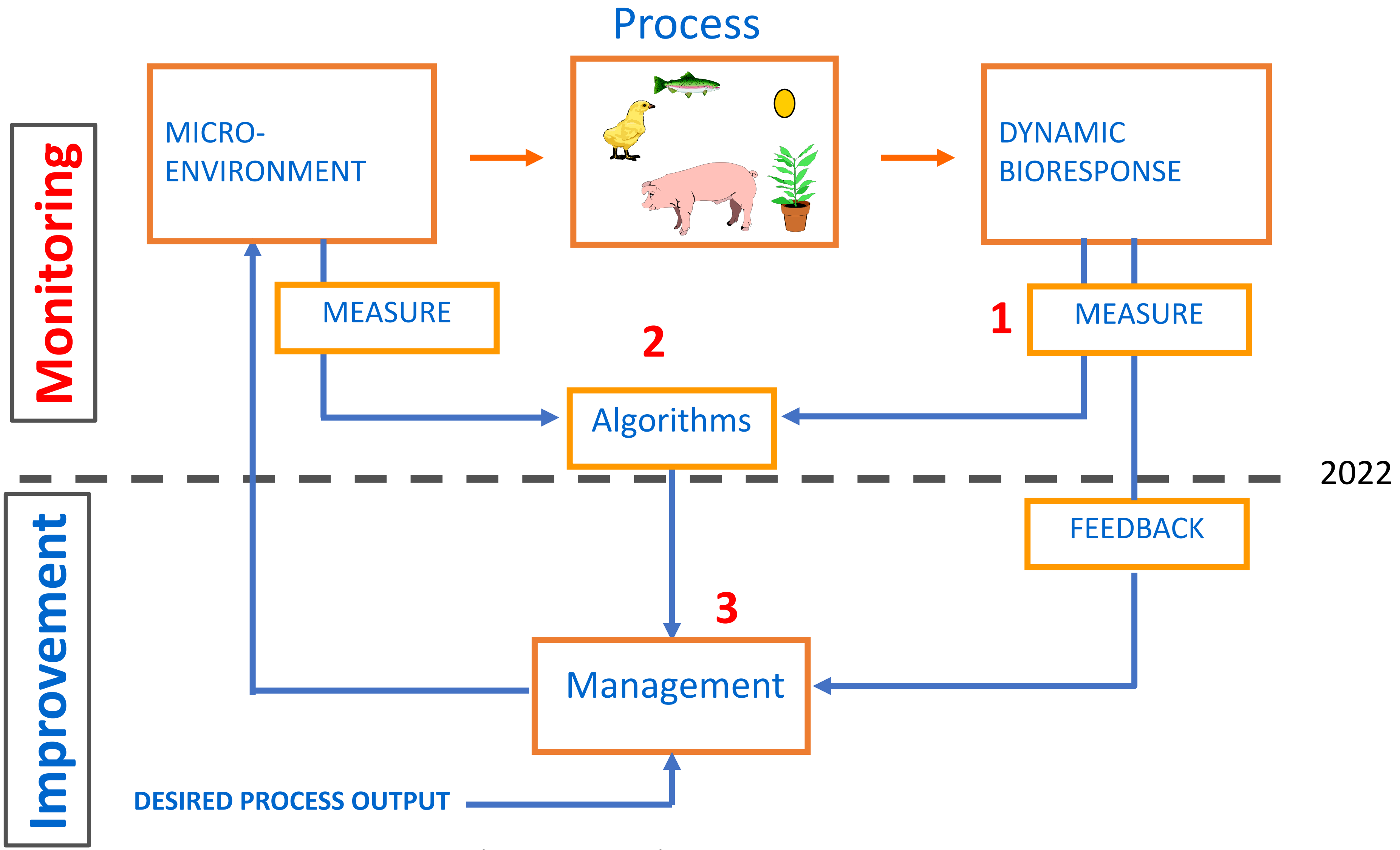
FEEDBACK

3

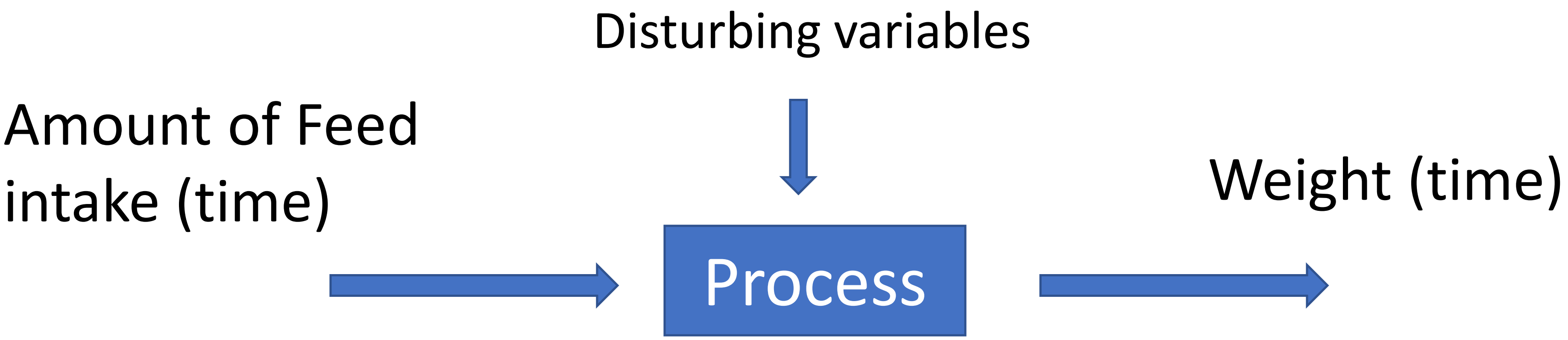
Desired direction

**Prediction-based controller**

# Model-based Monitoring/Improvement of Bioresponses (1991)



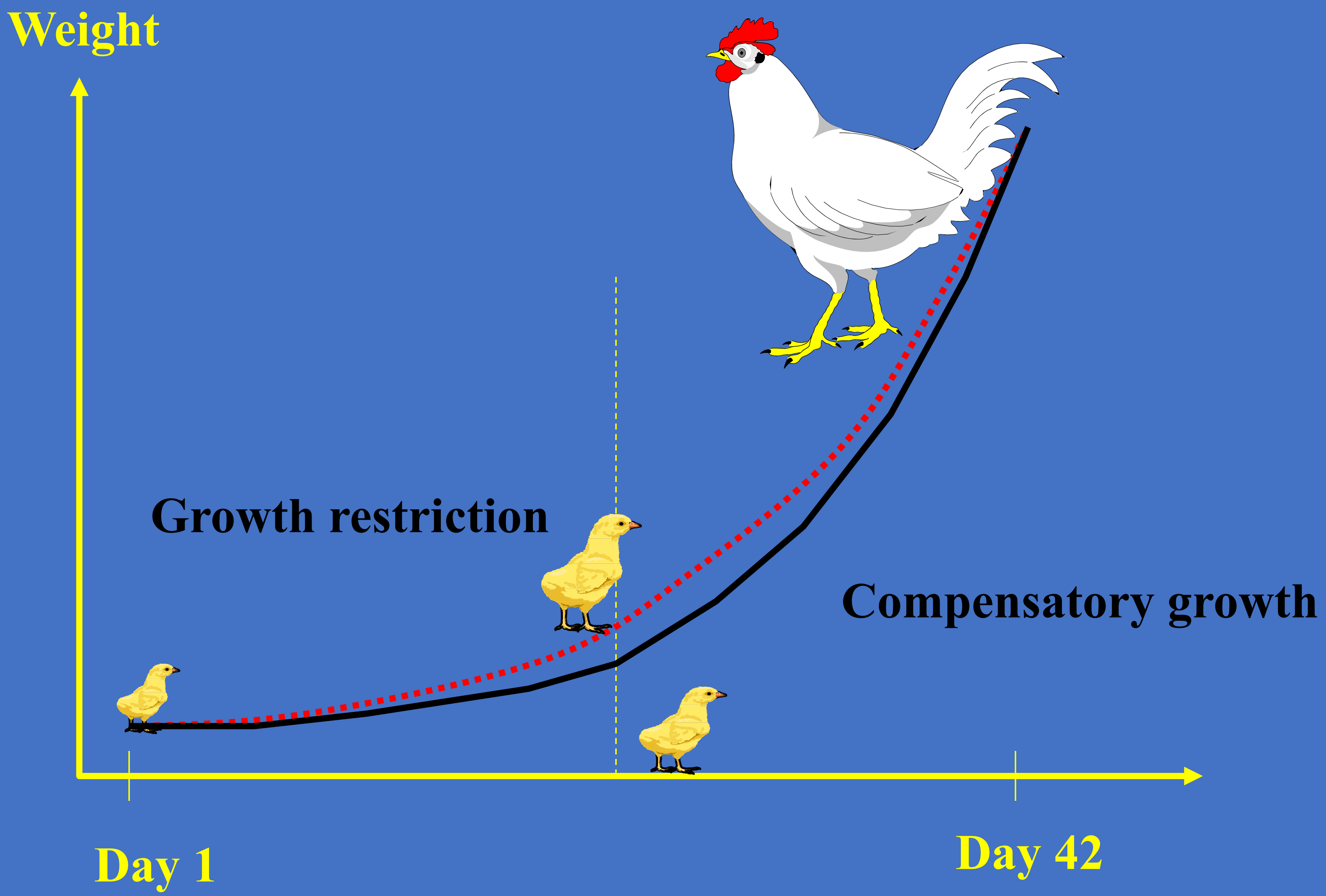
# Predictive based growth trajectory control of broilers

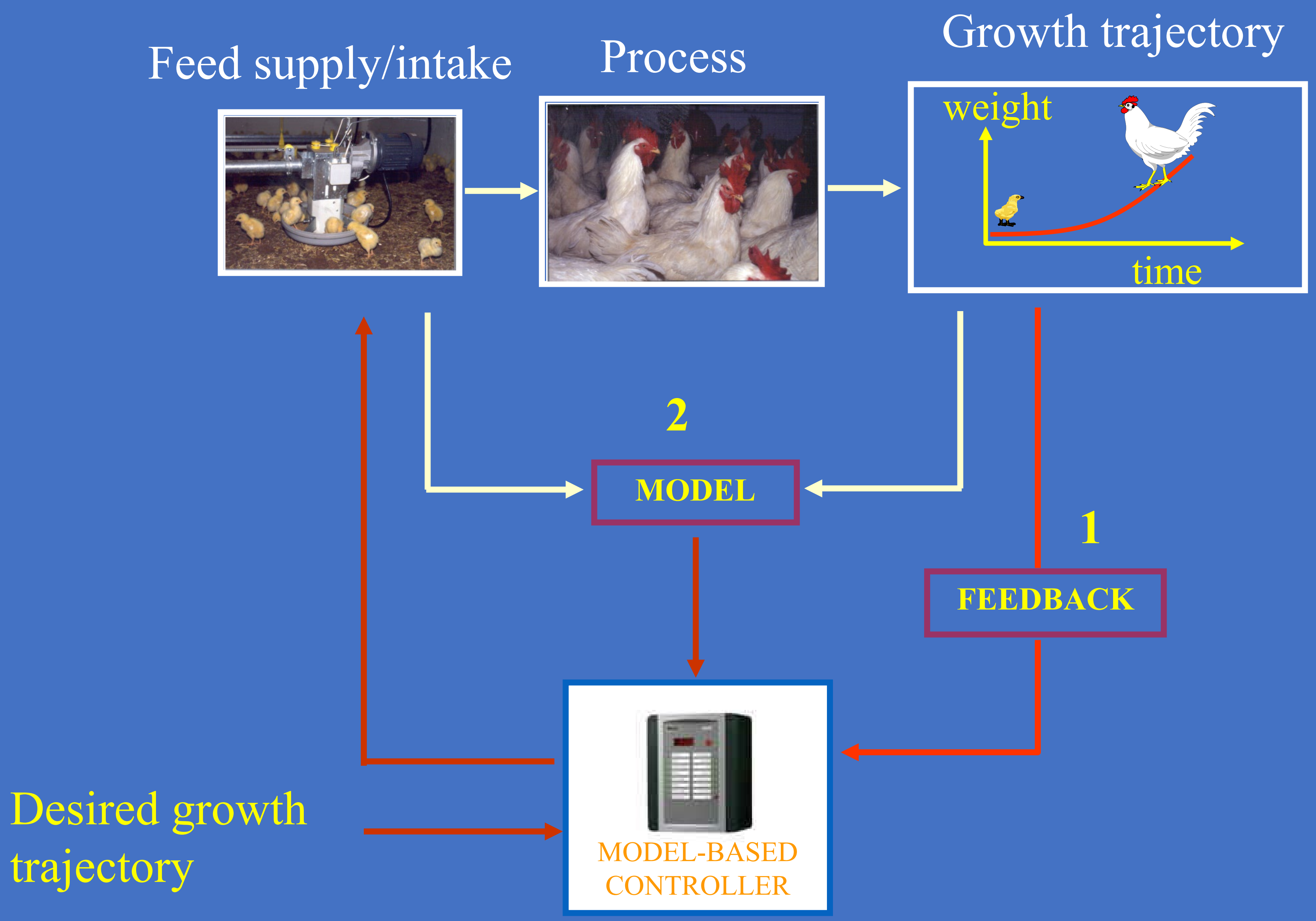


Possible solution:

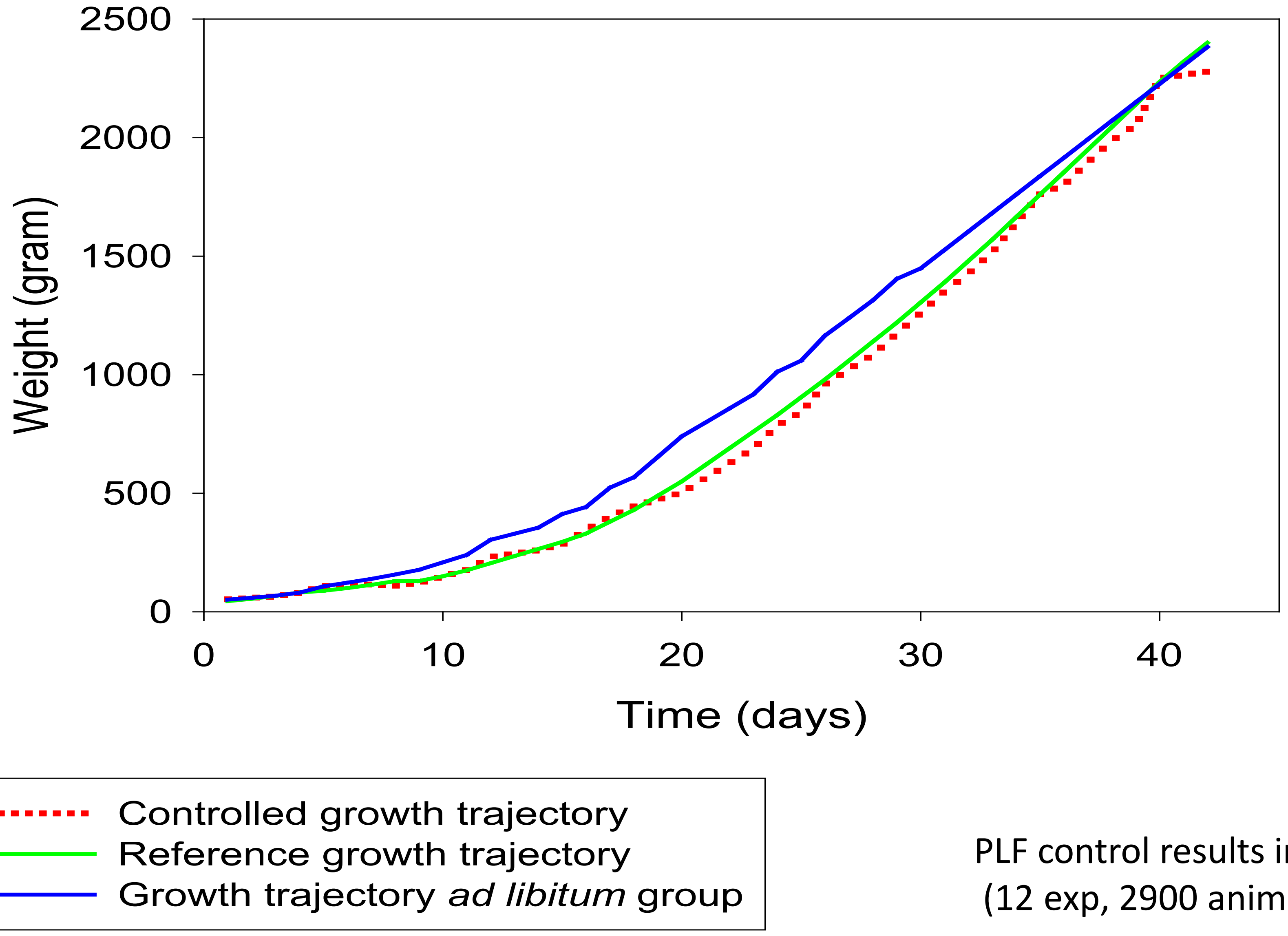
Growth restriction + compensatory growth

= Predictive-based growth control




Feed supply/intake
Process
Growth trajectory
weight
time
2
MODEL
1
FEEDBACK
MODEL-BASED
CONTROLLER
Desired growth
trajectory

# Results




PLF control results in reduction of mortality
(12 exp, 2900 animals each) of 4%

# Benefit of growth trajectory control (Broilers)

| Experiment | MRE | Mortality after week 1 | |
|---|---|---|---|
| | | Ad libitum group | Managed group |
| EXP1 | 3.7%,  4.1% | 5.0%, 6.0% | 0.0%, 3.0% |
| EXP2 | 4.5%,  5.3% | 3.3%, 3.0% | 1.1%, 1.3% |
| EXP3 | 4.0% | 4.1% | 2.1% |

Accuracy of continuous **InfectAlert algorithm** for humans

| | Specificity | Sensitivity | Sensitivity | Sensitivity | Accuracy |
|---|---|---|---|---|---|
| | | Covid-19 infections | Non-Covid infections | All | |
| **Focus on detection** | 80% | 75% | 90% | 79% | 80% |
| **Reducing false positives II** | 93% | 71% | 80% | 74% | 85% |

BioRICS
Biological Responses In Complex Systems

# Conclusions

- We need more animal product with less feed intake -> less manure, less emissions, less infections, better animal welfare.
- PLF offers *automated continuous real-time objective measurements*
- *PLF technology will allow to monitor all terms in the metabolic energy equation in real-time.*
- *Real-time monitoring & news sensors* allow improved *control to optimize* each term
- The *experts remains crucial* while PLF brings objectively measurements
- Implementation of PLF needs real *collaboration* between industry (hardware development, expertise), researchers, farmers and other stakeholders (policy makers, farmers organisations, …).

Thanks for your attention.

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